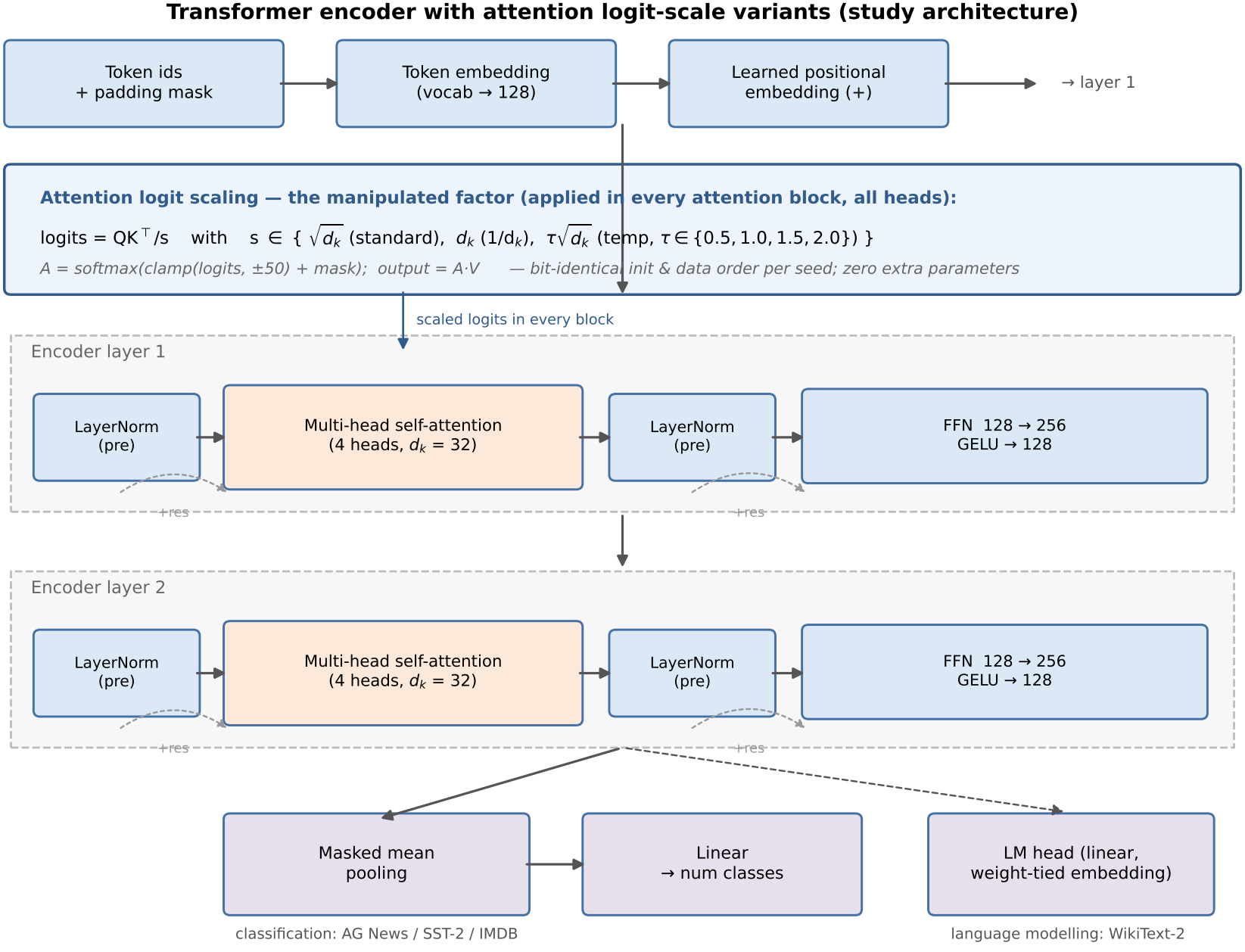
Transformer encoder with attention logit-scale variants (study architecture)
Token ids
+ padding mask
Token embedding
(vocab → 128)
Learned positional
embedding (+)
→ layer 1
Attention logit scaling — the manipulated factor (applied in every attention block, all heads):
logits = QK⊤/s with s ∈ { √d_k (standard), d_k (1/d_k), τ√d_k (temp, τ∈{0.5, 1.0, 1.5, 2.0}) }
A = softmax(clamp(logits, ±50) + mask); output = A·V — bit-identical init & data order per seed; zero extra parameters
scaled logits in every block
Encoder layer 1
LayerNorm
(pre)
Multi-head self-attention
(4 heads, d_k = 32)
LayerNorm
(pre)
FFN 128 → 256
GELU → 128
+res
+res
Encoder layer 2
LayerNorm
(pre)
Multi-head self-attention
(4 heads, d_k = 32)
LayerNorm
(pre)
FFN 128 → 256
GELU → 128
+res
+res
Masked mean
pooling
Linear
→ num classes
LM head (linear,
weight-tied embedding)
classification: AG News / SST-2 / IMDB
language modelling: WikiText-2